

Yuxin Shen

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EXPERIENCE	Research Associate, University of Cambridge, UK Sep 2026 – present
	Postdoc Research Associate in Dr Susanne Bornelöv's lab Working on developing better methods to train and interpret deep learning models for codon usage and gene expression control.
EDUCATION	University of Edinburgh, School of Biological Sciences, Edinburgh, UK
	PhD student in Biological Science (with Internship) Oct 2022 – present - Thesis: Machine learning for sequence-to-expression modelling and optimisation - Working on computational biology, applying machine learning to the problems in biological systems - Funded by BBSRC EASTBIO DTP
	Imperial College London, Department of Chemical Engineering, London, UK
	MSc in Advanced Chemical Engineering Oct 2020 – Sep 2021 - Master of Science with Distinction (Grade: 76.26/100) - Core Modules: Modelling of Biological Systems, Advanced Bioprocess Engineering, Advanced Process Operations, Dynamical Systems in Chemical Engineering, Biochemical Engineering, Advanced Process Optimisation - Master's thesis: Surrogate methods in model-based design of experiments for parameter precision
SCIENTIFIC RESEARCH EXPERIENCES	Fudan University, Department of Chemistry, Shanghai, China
	BSc in Chemistry Sep 2016 – Jul 2020 - Core Modules: Physical Chemistry, Organic Chemistry, Inorganic Chemistry, Analytical Chemistry, Introduction to Modern Biological Science, Topics in Organic Chemistry, and related experiments - Selected for Summer Programme in Chemical Biology at University of Hong Kong
	PhD Research, School of Biological Science, University of Edinburgh Oct 2022 – present
	Supervisor: Prof. Diego Oyarzún and Prof. Grzegorz Kudła
	Machine learning for sequence-to-expression mapping - Integrated biological domain knowledge into feature engineering for machine learning models. - Worked on both supervised learning (including RF, MLP, CNN, GNN and DNA large language models) and unsupervised learning for sequence-to-expression modelling. - Built a machine learning model for membrane protein expression with high prediction accuracy and explainability validated in wet lab experiments. - Created an active learning pipeline for DNA sequence generation and selection for protein expression optimization.
	Hybrid modelling of whole-cell ODE and ML - Conducted simulation and parameter estimation of a whole-cell ODE model using Julia. - Built machine learning models based on learned mechanistic parameters for growth rate prediction.
	MSc Research, Department of Chemical Engineering, Imperial College London Nov 2020 – Sep 2021
	Supervisor: Prof. Benoit Chachuat
RESEARCH OUTPUT	A comparative study of surrogate methods in model-based design of experiments - Applied different surrogates (simple neural network and Bayesian neural network) to approximate chemical reaction model sensitivities. - Conducted model-based design of experiments based on approximate sensitivities.
	1. Shen, Y., Kudla, G. and Oyarzún, D.A., 2025. Improving the generalization of protein expression models with mechanistic sequence information. <i>Nucleic Acids Research</i> , 53(3), gkaf020.
	2. Shen, Y., Kudla, G. and Oyarzún, D.A., 2025. Optimization of regulatory DNA with active learning. <i>Computational and Structural Biotechnology Journal</i> , 27, pp.4384-4392.
	3. Shen, Y., Underhill, J., Mulholland, A.J., Oyarzún, D.A. and Curnow, P., 2025. Effective sequence-to-expression prediction for membrane proteins using machine learning and computational protein design. <i>bioRxiv</i> , pp.2025-09. (Forthcoming in <i>ACS Synthetic Biology</i>)

4. Çubuk, H., Plech, M., Aslanzadeh, V., Zikanova, M., Skopova, V., Kmoch, S., Shen, Y., Marsh, J.A. and Kudla, G., 2025. Mechanistic Modelling of Recessive Disease through Allelic Integration of Variant Effects. *Cell Systems*, 17(6).
5. Hiropedi, A., Shen, Y. and Oyarzún, D.A., 2026. ezSTEP: a no-code tool for sequence-to-expression prediction in synthetic biology. (Forthcoming in *CIBB 2025, Lecture Notes in Computer Science*.)

INTERNSHIP EXPERIENCES

Informatics (Machine learning) Intern

Etcmby, Oxfordshire, UK

May 2025 – Jul 2025

- Built end-to-end bioinformatics workflows for large-scale immune repertoire data, including dataset construction, statistical hypothesis testing and comprehensive analysis and visualization
- Developed and implemented a pipeline of Large Language Models (LLMs) and biological domain knowledge for antibody and TCR sequence design.
- Fine-tuned a DNA LLM and designed antibody and TCR sequences with improved expression validated in wet-lab experiments.

Formulation R&D Technologist

Research and Development, Unilever (China) Shanghai Branch

Jan 2022 – May 2022

- Worked on product formulation development in laundry products for Homecare sector.
- Focused on new functions and sustainable solutions in liquid/concentrated laundry products.

Bioprocess Engineering Intern

Pall Corporation, Danaher Life Science Early Career Program

Sep 2021 – Dec 2021

- Assisted upstream bioprocess design for gene therapy and monoclonal antibody production.
- Designed single-use systems for bioreactors and the filtration processes.

Bioanalysis Intern

Industrial Research Institute, Shanghai Fosun Pharmaceutical (Group) Co., Ltd.

Dec 2019 – Feb 2020

- Assisted the PK/PD bioanalysis of the APIs for the generic drugs.
- Reviewed LC-MS methodology validation reports and bioanalytical reports.

HONORS & AWARDS

BBSRC EASTBIO PhD studentship (4-year full scholarship)

Oct 2022

Graduate Scholarship of Department of Chemistry

Jun 2020

Professional Scholarship of Department of Chemistry

Oct 2018 & Oct 2017

Outstanding Student Scholarship of Fudan University

Oct 2019 & Oct 2018 & Oct 2017

First Prize in Chinese Chemistry Olympiad in Shanghai

Oct 2015

PROGRAMMING SKILLS

Programming: Python (software development and deep learning), Julia, R, Linux/Bash Scripting.

- Python Packages for Scientific Computing and Machine Learning: NumPy, Pandas, SciPy, NetworkX, Matplotlib, Seaborn, Scikit-Learn, PyTorch, PyTorch Geometric, Transformers (HuggingFace), Tensorflow, JAX
- Julia Packages: DifferentialEquations, DiffEqFlux, Optimization, Catalyst
- Git, CI/CD workflows, Unit Testing, Containerization (Docker), High-Performance Computing (Slurm)

AI, Engineering Biology and Beyond Hackathon - Prize Winner

Bayesian Optimization Hackathon

Deep Learning for Healthcare Specialization Certificate on Coursera

Deep Learning Specialization Certificate on Coursera

Machine Learning Specialization Certificate on Coursera

OTHER EXPERIENCES

Alan Turing Institute Interest Group seminar series: Co-organizer

Sep 2024 – 202606

- Co-organize the Data-Centric Biological Design & Engineering Interest Group seminar series.
- Design, maintain, and update the seminar website using a GitHub-based static site (al-folio/Jekyll), including content layout, navigation, and styling.
- Lead speaker outreach, scheduling, and seminar coordination.
- Manage online dissemination and visibility of the seminar series through website updates and digital advertising.

BBSRC EASTBIO DTP Student Representative

Oct 2022 – Sep 2024

EASTBIO student representative for College of Science and Engineering, University of Edinburgh

- Organized panel and poster sessions in Annual EASTBIO Symposiums.
- Worked on Q&A information sessions for new students and potential applicants.

EASTBIO lead student representative on industrial engagement committee

- Worked on maintaining the network between industrial partners and PhD students.

**ATTENDED
CONFERENCES &
WORKSHOPS**

AI, Engineering Biology and Beyond 2026: Hackathon Prize Winner and Poster presentation	Jan 2026
Synthetic Biology UK 2025: Poster presentation	Nov 2025
12th BioProNET2 Annual Meeting: Poster presentation	Oct 2025
Computational Intelligence methods for Bioinformatics and Biostatistics 2025: Flash talk	Sep 2025
Bioprocessing Summit Europe 2025: Speaker	Mar 2025
Synthetic Biology UK 2024: Oral Communication	Nov 2024
Protein & Antibody Engineering Summit (PEGS) Europe 2024: Poster presentation	Nov 2024
Synthetic Biology for Health and Sustainability: Short talk	Oct 2024
EASTBIO Annual Symposium 2024: Speaker, Student representative and Co-Organizer	Jun 2024
2nd Workshop on Codon Usage: Poster presentation	May 2024
ChemEngDayUK 2024: Flash talk and poster presentation	Apr 2024
EMBL Symposium AI and biology: Flash talk and poster presentation	Mar 2024
BBSRC Inter-DTP Module in Software Engineering	Feb 2024
Synthetic Biology UK 2023: Poster presentation	Nov 2023
ELLIS Summer School on Machine Learning for Healthcare and Biology 2023	Jun 2023
EASTBIO Annual Symposium 2023: Student representative and Co-Organizer	Jun 2023
AI, Engineering Biology & Beyond - Turing Institute Workshop: Student Volunteer	Mar 2023
Host of EASTBIO Thematic Meeting “Computational tools for integrated Omics”	Feb 2023